A Conceptual Design of Kanji Mobile Application with Augmented Reality Technology for Beginner

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***Abstract*— Japanese language is the most popular in Thailand. Thai-Nichi Institute of Technology has Japanese language in undergraduate program as the require subjects and elective subject. The most difficult aspects of learning Japanese is Kanji. The aim of this research were as follow (1) to find out the problems and learning behaviors of Kanji learner, (2) to design Kanji mobile application with Augmented Reality technology. The populations were TNI students. Sampling of this research were 141 TNI students. The instrument of this research was questionnaire to collect data. The result was found were not able to read kanji, not able to write kanji by stroke order, able to guess the meaning but not able to read and not able to know meaning of kanji. In learning behaviors aspect was found the most learning behaviors were practice of writing often in paper or in the air, learning through printing media, learning through the game and learning through entertainment media. In the term of a conceptual design that selected Augmented Reality to resolve Kanji learning problem. And the color theme of an application designed by used blue tone that blue can help to focusing and to keep user engaged that color can be very effective in learning and educational. And the user interface designed by Flat design that benefit majority of Flat design were easy to use, beautiful and responsive.**

Keywords—kanji, mobile application, augmented reality technology

I. INTRODUCTION

The second language is the most important to communication with foreigner for exchange education, technology, economy, etc. The office of the basic education commission assigned English as second language that student must learn in basic education. But Japanese language is very popular too. Found 465 Japanese institutions in Thailand which consist of 129,616 Japanese learners. Because of the needs for work and study aboard. Japanese learners have increased dramatically [1] [2].

The samples in this research was TNI students (TNI stands for Thai-Nichi Institute Technology). Thai-Nichi Institute of Technology inserted Japanese language to undergraduate program as the require subjects and elective subject [3]. TNI students start to learn Japanese language from their first year. After completing their required and elective subjects in Japanese. The TNI students should be able to master their capabilities of Japanese language by passing level N4 of JLPT (JLPT stands for The Japanese-Language Proficiency Test) or in the other qualified test which is similar to JLPT [4]. The most difficult aspects of learning Japanese is Kanji [5] [6] [7]. From the result of research was found the students' age 18-19 years old who was born in 1998-1999 that was Generation Z [8].

Generation Z is Digital Natives have grown up in the Information Age and while comfortable with technology. Have used the Internet since a young age, and they are generally comfortable with technology and with interacting on social media. Generation Z uses different platforms for different activities such as Instagram, Facebook, Twitter. According to a study recently conducted by Response Media. From the characteristic of Generation Z, Augmented Reality Technology suitable to the age and generation of the students [8] [9]. Tanya and Nuanlaor's research were found learning and teaching with Augmented Reality Technology made the learner satisfaction in high level that the learner was fun excited and happy [10].

Augmented Reality (AR) allows learners to see the real world and have total immersion experiences. Which can applied to computers, tablets, and smartphones. Augmented Reality affords the ability to overlay images, text, video, audio, 3D models and motion pictures components onto existing images or space. Augmented Reality has the potential to change education to become more efficient in the same way that computers and internet. [11] [12].

The aim of this research were as follows:

1. To find out the problems and learning behaviors of the students who learning Kanji in beginner level.

2. To design Kanji mobile application with Augmented Reality technology.

II. USER REQUIREMENTS AND DATA GATHERING

A user requirement is the most important because that can specify what the application should be or able to do. The populations were TNI students. Sampling of this research were 141 TNI students derived through Simple Random Sampling technique. The instrument of this research was questionnaire to collect data. It was conducted through Google form. The questions were divided into 3 main sections:

Section 1: General personal information was the checklist question. The question asked six questions as a (1) gender, (2)

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year, (3) faculty, (4) age, (5) how long have you studied in Japanese? and (6) did you take the JLPT exam?

Section 2: Kanji learning problems and behavior was the checklist and rank order question.

Section 3: The current use of Kanji mobile application was checklist and open-ended question.

Data analysis was made by using various kinds of statistic such as percentage, average and frequency. The questionnaire results shown in Tables I to IV.

A. General personal information

TABLE I. GENERAL PERSONAL INFORMATION

General personal information		
No.	***Gender***	***Percent***
1	Male	66.7
2	Female	32.6
3	Other	0.7
No.	***Year***	***Percent***
1	Year 1	95.0
2	Year 2	0.7
3	Year 3	0.7
4	Year 4	3.5
No.	***Faculty***	***Percent***
1	Faculty of Information Technology	97.9
2	Faculty of Business Administration	2.1
3	Faculty of Engineering	0.0
No.	***Age***	***Percent***
1	Less than 18 years old	3.5
2	18 years old	56.7
3	19 years old	28.4
4	20 years old	3.5
5	21 years old	3.5
6	22 years old	2.8
7	More than 22 years old	1.4
No.	***How long have you studied Japanese?***	***Percent***
1	Never	69.5
2	3-6 months	8.5
3	6 – 12 months	4.5
4	13-24 months	3.5
5	More than 24 months	14.2
No.	***Did you take the JLPT exam?***	***Percent***
1	Never	81.6
2	Not pass	9.2
3	Passed N1	0.7
4	Passed N2	0.0
5	Passed N3	1.4
6	Passed N4	2.1
7	Passed N5	5.0

According to the Table I, the results show that the majority of the students were male 66.7%, first-year student of Information Technology faculty 97.9%, 18 years old 56.7%, Non-kanji or Japanese language background 69.5% and never taken any level of JLPT exam 81.6%.

B. Kanji learning problems and learning behaviors

From Table II the students were arrange Kanji learning problems in ascending order. The first problems were not able to read kanji, second were not able to write kanji by stroke order. The third and fourth problems were able to guess the meaning but not able to read and not able to know meaning of kanji.

TABLE II. KANJI LEARNING PROBLEMS

Arrange your Kanji learning problems in ascending order.						
No.	***Questions***	***1st***	***2nd***	***3rd***	***4th***	***5th***
1	Not able to read Kanji	58	29	12	14	28
2	Not able to write Kanji	25	14	55	24	23
3	Not able to write Kanji with stroke order.	13	28	18	55	27
4	Able to guess the meaning but not able to read.	19	20	25	24	53
5	Not able to know the meaning of Kanji.	26	50	31	24	10

[a.] Ranking Question

TABLE III. KANJI LEARNING BEHAVIORS

Kanji learning behaviors		
No.	***Questions***	***Percent***
1	Practice of writing often in paper or in the air.	56.7
2	Learning through printing media such as comic, magazine and novel.	43.3
3	Learning through the game.	43.3
4	Learning through entertainment media such as movie, drama, song and animation.	41.8
5	Practice often with Kanji exercise.	39.7
6	Remember in photographic memory	39.0
7	Ask the teacher or friend when have a question.	38.3
8	Learning through the application.	36.9
9	Learning through the website.	32.6
10	Remember Kanji from Hiragana or Katakana.	31.2
11	Learning through the flash card game or another game with friend.	27.0
12	Learning through the environment such as signboard and placard.	26.2
13	Use flash card.	24.8
14	Learning through the dictionary.	21.3
15	Remember by a group.	14.2

In learning behaviors aspect from table III was found the most learning behaviors were (1) practice of writing often in paper or in the air, (2) learning through printing media such as comic, magazine and novel, (3) learning through the game, and (4) learning through entertainment media such as movie, drama, song and animation.

C. The current use of Kanji mobile application

The result was shown in Table IV. the majority of the students never used Kanji mobile application 51.1% and used Kanji mobile application 48.9%. The majority of operating system was Android. In the term of mobile application function was found 3 functions that should able to do namely: stroke order animation, pronunciation and translate in Thai, Native speaker sound. The students gave opinion concerning about the Kanji mobile application can improve Kanji skill better in high level 41.1%, average 39.7%, highest level 8.5%.

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TABLE IV. KANJI MOBILE APPLICATION SHOULD ABLE TO DO

Kanji mobile application function should able to do		
No.	***Questions***	***Percent***
1	Stoke order animation	81.6
2	Pronunciation and translate in Thai	73.8
3	Native speaker sound	70.2
4	Illustration in Motion picture	66.0
5	Classified by a group	65.2
6	Sample sentence	63.8
7	Classified by JLPT level	63.1
8	Illustration in still image	61.7
9	Related vocabulary	61.0
10	Game	55.3

D. Questionnaire result conclusion and discussion

As a whole, it was found from this questionnaire results that the students was male, first-year student in Information Technology faculty, 18 years old, No Japanese language background and never taken any level of JLPT test. In other words, they were beginner level of Japanese learning. The majority of Kanji learning problem were 4 problems namely: (1) not able to read Kanji, (2) not able to write Kanji, (3) not able to write Kanji with stroke order, and (4) able to guess the meaning but not able to read and not able to know the meaning of Kanji.

The students were born in 1998-1999 that was Generation Z. Most of Generation Z have used the Internet since a young age, and they are generally comfortable with technology and with interacting on social media. Generation Z uses different platforms for different activities such as Instagram, Facebook, Twitter. According to a study recently conducted by Response Media [9].

Form above the researcher selected Augmented Reality Technology to represent new learning material of Kanji in mobile application platform. To solve Kanji learning problem and suitable for the ages and generation of the students. AR technology can be utilized through a variety of mediums including desktops, mobile devices, and smartphone. The technology is portable and adaptable to a variety of scenarios [11]. AR can be used to enhance content and instruction within the traditional classroom, supplement instruction in the special education classroom, extend content into the world outside the classroom, and be combined with other technologies to enrich their individual.

III. CONCEPTUAL DESIGN OF KANJI MOBILE APPLICATION

A conceptual design was based form user requirement in "user requirements and data gathering" section that included contents, characters design, storyboard, user interface, framework to understand in the whole process.

A. A conceptual Framework

The conceptual framework of Kanji mobile application divided into 3 main parts were Application, Augmented Reality SDK and 3D Models. In the term of application use Unity to created application for Android OS that because from the result of user requirement in "user requirements and data gathering" section. The majority of the students used Android OS.

Fig. 1 is shown the core component of Augmented Reality SDK that consists of three steps: step 1 the camera captures the preview frame and pass to the image tracker, step 2 the pixel format converter converts the default format of the camera to a format suitable for OpenGL ES rendering and tracking, step 3 the tracker detects and tracks real world objects in camera preview frame using image recognition algorithms. The results are stored in state object which is passed to the video background renderer, and step 4 the renderer renders the camera image stored in state object and augments it on to the camera screen [13].

The device database or dataset store Image Target, Multi Targets, and Cylinder targets on the device. Created using

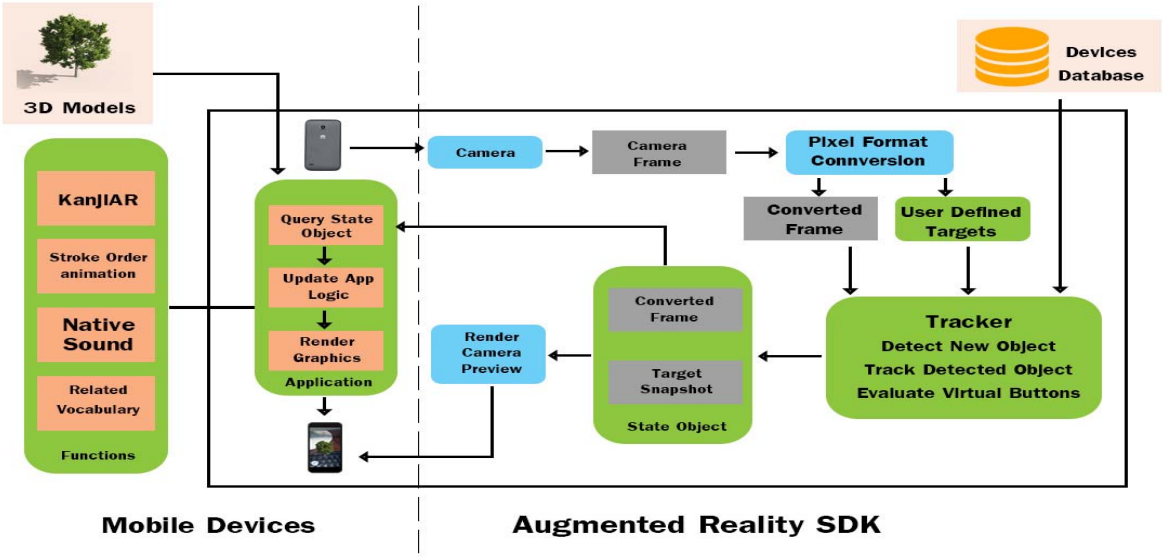

Fig 1. Conceptual Framework

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Target Manager. It contains an XML configuration file and binary .dat file [14].

The 3D models created based from the character designs and storyboard in section B by using MAYA, Unity Game Engine and converted into .obj file format.

B. Contents

The main contents of Kanji Mobile Application were Kanji alphabets in JLPT N5 level that beginner need to know. Actually Kanji in JLPT N5 were more than eighty words [15]. But in this research selected to design for prototype six characters namely: 川, 木, 山, 雨, 火, 月. to created 3D models.

1) Characters Design

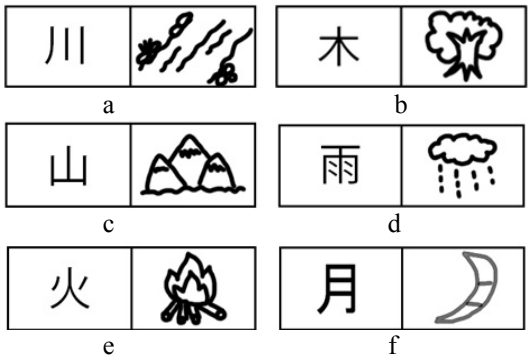


Fig. 2. Character design

a) Fig. 2 (a): 川 pronounced "kawa" is mean river. Have 3 strokes. The characters designed was similar to a river or stream according to the meaning of this word.

b) Fig. 2 (b): 木 pronounced “ki” is mean tree, wood, timber, and shrub. This kanji have four strokes. The characters designed was similar to a tree have three roots on bottom according to the meaning of this word.

c) Fig. 2 (c): 山 pronounced “yama” is mean mountain. This kanji have three strokes. The characters design was similar to three mountains with snow capped on a mountain peak according to the meaning of this word.

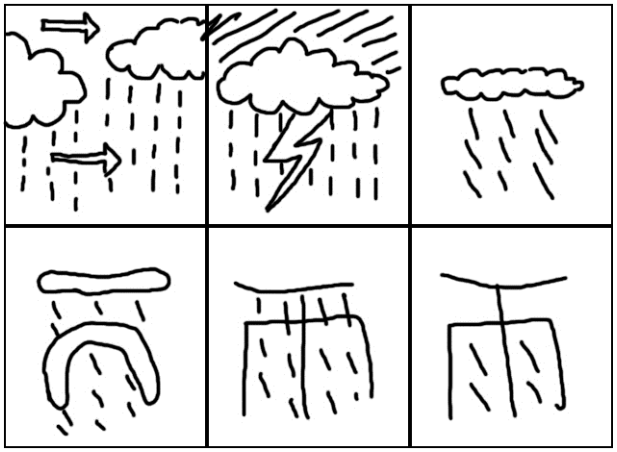
d) Fig. 2 (d): 雨 pronounced “ame” is mean rain. This kanji have eigth strokes. The characters design was similar to clouds with rain according to the meaning of this word.

e) Fig. 2 (e): 火 pronounced “hi” is mean fire, flames, light, heat. This kanji have four strokes. The characters design was similar to bonfire according to the meaning of this word.

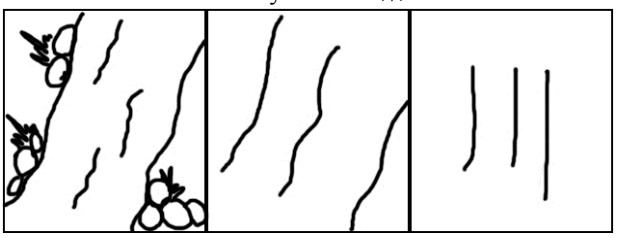
f) Fig. 2 (f): 月 pronounced “tsuki” is mean moon,. This kanji have four strokes. The characters design was similar to crescent moon according to the meaning of this word.

2) Story board

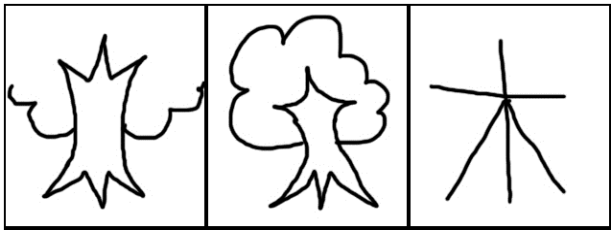
A storyboard is a graphic organizer in the form of illustrations or images displayed in sequence for the purpose of pre-visualizing a motion picture, animation, motion graphic or interactive media sequence. In this section was shown storyboard of Kanji. From graphic pictogram to Kanji characters step by step as shown in Fig. 3



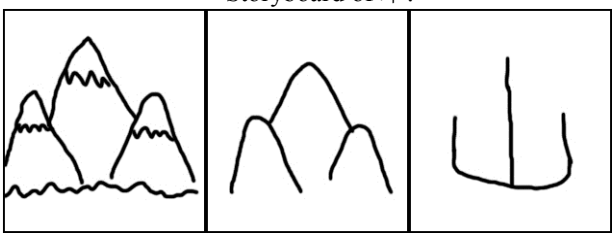
Storyboard of 雨.



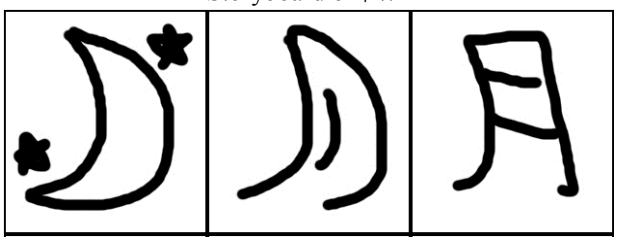
Storyboard of 川.



Storyboard of 木.



Storyboard of 山.



Storyboard of 月.

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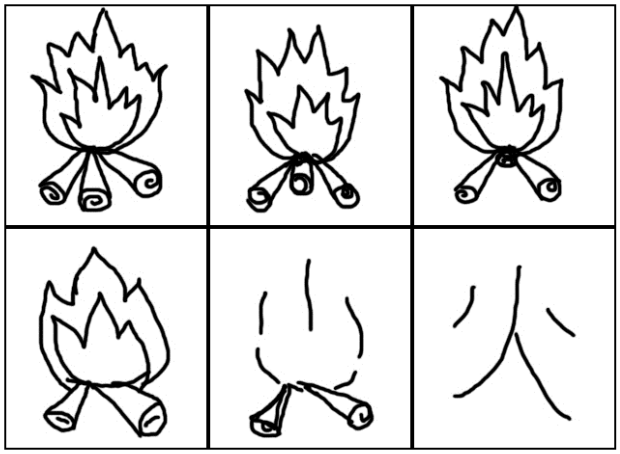

Storyboard of 火.

Fig. 3. Storyboard design

C. User Interface (UI)

A user interface is important that to show and interactive with users. User interface is the front-end presentation layer of software system. Can be in form of text, image, and video. In the terms of user interface that to showed user interface of Kanji mobile application.

1) Color theme

Fig. 4 was shown the user interface design of splash screen, main menu, Kanji detail, Kanji AR, Kanji stroke order animation AR. All interface to designed in blue tone that blue was cool undertone. Because blue can help to focusing and to keep user engaged that color can be very effective in learning and educational [16].

2) User Interface theme

User Interface theme in this research designed in the concept of flat design. Flat desing is a style of interface design emphasizing minimum use of stylistic elements that give the illusion of three dimensions (such as the use of drop shadows, gradients or textures) and is focused on a minimalist use of simple elements, typography and flat colors. Benefit majority of Flat design were easy to use, beautiful, responsive [17].

3) Features

The main features of the Kanji mobile application from this research was Kanji characters display in three-dimensions animation with Augmented Reality Technology. And another features were stroke order animation, pronunciation and translate in Thai, Native speaker sound. Fig. 4 (a) was splash screen that showed title and logo. Fig. 4 (b) was main menu shown Kanji characters that each button can link to Kanji detail interface see Fig. 4 (c). Kanji detail interface showed Kanji charecters, related kanji translate in Thai, button to play native speaker sound, and play button to link to Kanji AR interface. To show Kanji 3D animation that developed from characters design and storyboard in above section with Augmented Reality Technology as shown in Fig. 4 (d and e).

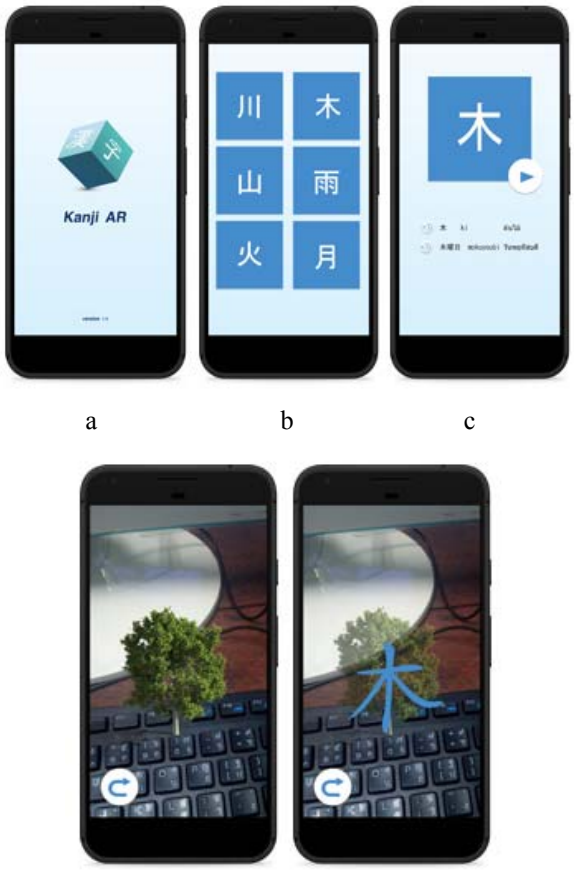

Fig. 4. (a) Splash screen, (b) Main menu, (c) Kanji detail, (d) Kanji AR and (e) Kanji stroke order

IV. Conclusions

In this research presented a conceptual design of Kanji mobile application with Augmented Reality technology for beginner. The aim of this conceptual design is to specify and visualize in what way the design of Kanji mobile application is based on the user requirements.

The students was TNI students 18-19 years old, a first-year student, was born in generation z, no Japanese background, never taken any level of JLPT test, used Android OS. In the term of Kanji learning problem that the result was found the first problems were not able to read kanji, second were not able to write kanji by stroke order. The third and fourth problems were able to guess the meaning but not able to read and not able to know meaning of kanji. In learning behaviors aspect was found the most learning behaviors were practice of writing often in paper or in the air. Second were learning through printing media such as comic, magazine and novel. The third to fourth were Learning through the game and Learning through entertainment media such as movie, drama, song and

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animation. In the students opinion about the Kanji mobile application can improve Kanji skill better in high level 41.1%. The function should able to do was found: stroke order animation, pronunciation - translate in Thai and Native speaker sound.

In the term of conceptual design, the color theme of an application designed by used blue tone that blue can help to focusing and to keep user engaged that Color can be very effective in learning and educational. And the user interface designed by Flat design that benefit majority of Flat design were easy to use, beautiful and responsive. In the term of application's features were Kanji characters display in thee-dimension animation with Augmented Reality Technology. And another features were stroke order animation, pronunciation - translate in Thai and Native speaker sound.

Augmented Reality Technology that suitable for the students. Because gave the new experience excited rousing the user. In education Augmented Reality Technology helps to enhance learning though motivation. To resolve Kanji learning problem AR is considered as a good tool.

In future work this conceptual design will be building and implement to the students.

ACKNOWLEDGMENT

The researcher would like to especially thank all teachers in Thai-Nichi Institute of Technology and the respondents for support.

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